

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Examination, 2021
CSE – 1201, Fundamentals of Programming

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) C supports five different basic data types. Expand & Explain. 5
 b) Write short notes on local, global and static variables. What are the features of static local and static global variables? 5
 c) Write down the definitions of Token, Keyword, and Identifier with proper examples. 4

2. a) Differentiate the behaviors of the following keyword using suitable code example: 4
 i. break ii. continue iii. case iv. default
 b) What will be the output for the following programs: 2×3=6

i) <pre>#include<stdio.h> int main () { int a = 10, b = 25; a = b++ + a++; printf("%d%d",a,b); b = ++b + ++a; printf("%d%d",a,b); }</pre>	ii) <pre>#include<stdio.h> void main() { int a = 0, b = 1; b = 5 2; printf("%d",b); a = b 2 1; printf("%d",a); }</pre>	iii) <pre>#include <stdio.h> int main(){ int* pc, c; c = 22; printf("Address of c: %p\n", &c); printf("Value of c: %d\n", c); pc = &c; printf("Address of pointer pc: %p\n", pc); printf("Content of pointer pc: %d\n", *pc); return 0; }</pre>
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 c) Explain the syntax of the switch case construct. Write a menu-driven 'C' program using a switch case to take 2 integers as input and print the sum, difference, and product depending upon the choice provided by the user. 4

3. a) What do you mean by operator precedence? What is the associativity of the arithmetic operator? 2+2
 b) Write the structures of the following statements: 3
 i. for ii. do-while
 c) How can you read and write two float values in a file? 2
 d) What is the output when the following code fragments are executed? 1+4

I) <pre>int A=1; int B=2; if((A==1) (B==2) && (B==0)) printf("This Exam was difficult"); else printf("This Exam was easy");</pre>	II) <pre>int n; printf("Enter the number of n:\n"); scanf("%d",&n); if (n<10) printf("Less than 10"); else if(n>5) printf("Greater than 5"); else printf("Not Inserting");</pre>
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Based on the question 3. d) II)

- i) What will the output of the fragment be if the interactive user enters the integer value 0?
- ii) What will the output of the fragment be if the interactive user enters the integer value 15?
- iii) What will the output of the fragment be if the interactive user enters the integer value 7?
- iv) What are the values for n that cause the output of the fragment above to be "Not inserting"?

4. a) Differentiate between the entry-controlled loop and exit-controlled loop. 3
 b) Write a loop statement that will show the following output. 5
- ```

1
2 1
3 2 1
4 3 2 1
5 4 3 2 1

```
- c) Write a function that displays a solid square of asterisk whose side is specified as an integer parameter side. For example, if the side is 4, the function displays: 3
- ```

****
****
****
****

```
- d) What are the differences between an else-if ladder and a switch case? 3
5. a) What, if anything, prints, when each of the following C statements, is performed? If the statement contains an error, describe the error and indicate how to correct it. Assume the following variable definitions: 4
- ```

char s1[50]= "FOELT", s2[50]= "CSE", s3[50];
i. printf("%s\n",strcpy(s3,s2));
ii. printf("%s\n",strcat(strcat(strcpy(s3,s1),"and"),s2));
iii. printf("%d\n",strlen(s1)+strlen(s2));
iv. printf("%d\n",strlen(s3));

```
- b) Write a program in C using pointer to search the smallest element in an array of numbers. 4  
 c) What is the difference between static memory allocation and dynamic memory allocation? 3  
 d) What are the differences between malloc () and calloc ()? Which one is more preferable to use? 2+1
6. a) Why string is called a character-type array? How can we write a program that will take "Albert" as input? 1+2  
 b) What is an array? How to declare an array? Write a program to find the average of four numbers using an array. 1+2+3  
 c) Write a for-loop statement to compute the result of the following sequence of integers and print it: 2.5×2
- ```

i) 1+2+3+4+...+n
ii) 12+22+32+...+n2

```
7. a) Define preprocessor directive. What will the preprocessor do for a program? 4
 b) Write a single statement to accomplish each of the following. Assume that each of these statements applies to the same program. 6
- Write a statement that opens file "previousAC.dat" for reading and assigns the returned file pointer to ofPtr.
 - Write a statement that opens file "transferAC.dat" for reading and assigns the return file pointer to tfPtr.
 - Write a statement that opens file "newAC.dat" for writing and assigns the returned file pointer to nfPtr.
 - Write a statement that reads a record from the file "previousAC.dat". The record consists of integer accountNum, string name, and floating point currentBalance.
 - Write a statement that reads a record from the file "transferAC.dat". The record consists of integer accountNum and floating point dollarAmount.
 - Write a statement that writes a record to the file "newAC.dat". The record consists of integer accountNum, string name, and floating point currentBalance.
- c) Write a C program to define Function-like-Macro to find the minimum of two numbers. 4

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1st Year 2nd Semester B.Sc. Examination, 2021
CSE – 1202, Digital Logic Design

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Explain the operation and prove the truth table of the following circuit. 5
- b) Describe Shift Register and Decade counter. 4
- c) Explain the working principal of S-R flip flop with NAND gate. 5
2. a) Simplify the expression: $\overline{(A+B)}.B.(A+\bar{C})$. 3
- b) Minimize the following function using Karnaugh map:
 $F(A, B, C, D) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(10, 13)$ 5
- c) Write the output expressions and draw the circuit diagram of a 3-bit comparator. 6
3. a) Design an OR gate using NAND gate. 2
- b) Draw the circuit diagram of a full adder using two half adders. 4
- c) Implement the function using an 8x1 Multiplexer. $F(A, B, C, D) = \sum m(0, 2, 4, 6, 8, 10, 11)$. 5
- d) Draw the block diagram of a 16x1 MUX. 3
4. a) Find the 10's complement of $(325.625)_{10}$. 2
- b) Implement a two input X-OR gate using only two input NAND gates. 3
- c) Convert the following expressions into standard SOP(Sum of Product) form: 4
 - i. $\overline{(A+B)} + C$
 - ii. $(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$
- d) Develop a truth table for the following SOP (Sum of Product) expression: 3
 $\bar{A}B + AB\bar{C} + \bar{A}\bar{C} + A\bar{B}C$
- e) Convert the binary code 101101 to Gray and vice-versa. 2
5. a) Implement the following functions using PLA: 6

$$F1(x, y, z) = \sum m(1, 2, 3, 6)$$

$$F2(x, y, z) = \sum m(0, 1, 5, 7)$$

$$F3(x, y, z) = \sum m(5, 6)$$

$$F4(x, y, z) = \sum m(1, 2, 3, 4, 7)$$
- b) Explain "Race Condition" in Flip-Flop with circuit diagram and truth table. 4
- c) Draw the block diagram and truth table of a BCD to 7 segment display system. 4
6. a) Minimize the following Boolean expressions: 3
 - i) $F(A, B, C, D) = \sum m(3, 4, 5, 7, 8, 9, 10)$ 4
 - ii) $F(A, B, C) = \sum m(3, 4, 5) + \sum d(0, 1, 6, 7)$ 2
- b) Define Propagation Delay and Noise Margin. 5
- c) Describe 8x1 MUX with Circuit diagram and output expression. 5
7. a) What is counter modulus? 2
- b) Design a 4-bit parallel adder. 3
- c) What is the difference between S-R flip-flop and J-K flip-flop? 1
- d) Draw the circuit diagram of J-K flip-flop with truth table. 4
- e) Prove the universality of NOR gate. 4

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PHY – 1203, Physics

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) What do you mean by entropy? Prove that entropy is constant in reversible process. 3
 b) What is adiabatic change? For adiabatic process prove that $PV^\gamma = \text{Constant}$. 7
 c) A motor car tire has a pressure of 2 atmospheres at 25°C. It is suddenly bursts, find the resulting temperature. ($\gamma = 1.4$) 4
2. a) Define: (i) Degrees of freedom (ii) 2nd law of thermodynamics. 3
 b) Find the equation of work done for expansion of gas in adiabatic and isothermal process. 7
 c) A Carnot engine whose $T_1 = 7^\circ\text{C}$ and efficiency of 50%. It is desire to increase of efficiency to 70%. By how many degrees should the temperature of the high be increased? 4
3. a) What Simple Harmonic Oscillation? Write down its characteristics. 3
 b) Show that maximum K.E and maximum potential energy is same and equal to total energy. 7
 c) A body is vibrating with simple harmonic motion of amplitude 15 cm and frequency 4 Hz. Compute (i) the maximum acceleration and velocity (ii) the acceleration and velocity when the displacement is 9 cm. 4
4. a) What is (i) Particle velocity (ii) damped harmonic oscillation (iii) reduce mass 3
 b) For damped harmonic oscillation prove that, $x = e^{-bt} A \cos(\omega t + \delta)$ 7
 c) The velocity of a simple harmonic wave is 30cm/s. At a time $t = 0$ the displacement of a particle is given by $= 4 \sin 2\pi \frac{x}{100}$. Find the equation for the displacement at a time $t = 2\text{s}$. 4
5. a) Define packing fraction. Calculate the packing fraction for Simple cubic crystal structure. 5
 b) What do you mean by the term miller indices? Show that the inter-planar distance of orthorhombic crystal is given by

$$d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$
 6
 c) Show that in a simple cubic lattice the separation between the successive lattice planes (100), (110) and (111) are in the ratio 1: 0.71: 0.58. 3
6. a) What are Newton's rings and discuss why the Centre of Newton's rings is always dark. 5
 b) In the phenomena of interference, show that the width of the bright and the dark fringe are always equal. 5
 c) In a Newton's ring experiment, the diameter of 15th ring was found to be 0.59 cm and that of the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, calculate the wavelength of light. 4
7. a) Define the terms: Polarization of light, Double refraction and Nicol prism. 3
 b) State and explain the Brewster's Law. From Brewster's law show that the polarizing angle of incidence, the reflected and refracted rays are perpendicular to each other. 7
 c) The refractive index of plastic is 1.25. Calculate the angle of refraction for a ray of light incident of polarizing angle. 4

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MATH – 1204, Method of integration, differential equations & Series

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Integrate the following integrals: 2×2
 - i) $\int \frac{x + \sin x}{1 + \cos x} dx$
 - ii) $\int \frac{dx}{\sin^{\frac{1}{2}} x \cos^{\frac{7}{2}} x}$
- b) Find the surface area of the solid generated by revolving the parabola $y^2 = 4ax$ between $x = 0$ and $x = a$ about x-axis. 4
- c) Evaluate (for any two) : 3×2
 - i) $\int \frac{\sqrt{x^2 - a^2}}{x^3} dx$
 - ii) $\int \frac{dx}{5 + 4 \sin 2x}$
 - iii) $\int \tan^{-1} \sqrt{x} dx$
2. a) Define Order and Degree of differential equation with an example. 2
- b) Form a differential equation representing the given family of curves by eliminating arbitrary constants a and b where $y = e^x(acosx + bsinx)$ 3
- c) Solve the following differential equations: 9
 - (i) $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$
 - (ii) $\cos^2 x \frac{dy}{dx} + y = \tan x$
 - (iii) $\sin^{-1} \left(\frac{dy}{dx} \right) = x + y$
3. a) Define linear differential equation. Solve the differential equation: 6

$$(1 - x^2) \frac{dy}{dx} - xy = 1$$
- b) Show that the necessary and sufficient condition for the first order ordinary differential equation $M dx + N dy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ 4
- c) Solve the differential equation: $(e^y + 1)\cos x dx + e^y \sin x dy = 0$ 4
4. a) Define convergency and divergency of a series. 2
- b) Test the following series for convergence: (i) $\sum_{n=1}^{\infty} \frac{1}{2n!}$ (ii) $\sum_{n=1}^{\infty} \sqrt[n]{\left(\frac{n^2+1}{2n^2+1}\right)^n}$ 6
- c) Apply Taylor's method to find $y(x)$ at the point $x = 1.1$ and $x = 1.2$ by solving $\frac{dy}{dx} = x^2 + y^2$ and $y(1) = 2.3$. 6

5. a) Show that the arc length of the Curve $y = \ln \sec x$ from $x = 0$ to $x = \frac{\pi}{3}$ is $\ln(2 + \sqrt{3})$ 5
- b) Using Euler's method to the ordinary differential equation $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$ for values at the point $x = 0$ to $x = 0.4$ taking $h = 0.1$ 4
- c) If, when the temperature of the air is 20°C , a certain substance cools from 100°C to 60°C in 10 minutes, find the temperature after 40 minutes. 5

6. a) The number of bacteria in a culture is known to increase at a rate proportional to the number of bacteria present at time t . A culture contains 10,000 bacteria initially. After an hour, the bacteria count is 25,000. Find the doubling period? 4
- b) Expand $\log(1+x)$ in power of x by Maclaurin theorem. 5
- c) Find the perimeter of the cardioid $r = 2(1 - \cos\theta)$ 5

7. a) Show that $y = (x^2 + c)e^{-x}$, where c is constant, is the general solution of $\frac{dy}{dx} + y = 2xe^{-x}$. Also find a particular solution for the condition $y(-1) = e + 3$ 5

- b) Find the differential equation whose solution is $y = ae^x + be^{-x} + c \cos x + d \sin x$ where a, b, c, d arbitrary constants. 4

- c) Evaluate the improper integral, $\int_{-\infty}^{\infty} \frac{x}{x^4 + 1} dx$ 5

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ENG – 1205, Developing English Language Skills

Total Marks: 70

Time: 3 Hours

Part A (Comprehension: 15 marks)

Read the following passage and answer the questions given below:

Good governance is the process of measuring how public institutions conduct public affairs and manage public resources and guarantee the realization of human rights in a manner essentially free of abuse and corruption and with due regard for the rule of law. Governance is "the process of decision-making and the process by which decisions are implemented (or not implemented)". Governance in this context can apply to corporate, international, national, or local governance as well as the interactions between other sectors of society.

The concept of "good governance" thus emerges as a model to compare ineffective economies or political bodies with viable economies and political bodies. The concept centres on the responsibility of governments and governing bodies to meet the needs of the masses as opposed to select groups in society. Because countries often described as "most successful" are liberal democratic states, concentrated in Europe and the Americas, good governance standards often measure other state institutions against these states. Aid organizations and the authorities of developed countries often will focus the meaning of "good governance" to a set of requirements that conform to the organization's agenda, making "good governance" imply many different things in many different contexts.

1. Answer the following questions: 5
 - a) What is the central idea of the given passage?
 - b) What are the factors that good governance includes?
 - c) What is the responsibility of the government?
 - d) What do the aid organizations and developed countries do?
 - e) What does good governance mean?
2. Write the meaning of the following words in English. 5
a) viable b) measure c) agenda d) emerge e) requirement.
3. Write the précis of the given passage. 5

Part B (Grammar: 25 marks)

4. Correct *any five* of the following sentences. 5
 - a) I went to bazaar with a view to buy some books.
 - b) Recently we are going to Dhaka.
 - c) My brother has returned home last week.
 - d) That he bought the books are known to me.
 - e) You're right to say so.
 - f) When the book first published?
 - g) Dickens is a populous novelist.

5. Transform *any five* of the following sentences as directed. 5
- I have never thought of going there. (Make it Complex)
 - To err is human. (Make it Negative)
 - Since the weather was misty, we cancelled our tour. (Make it Simple)
 - The scenery of St Martin is more charming than that of Kuakata. (Make it Positive Degree)
 - He helped me do the work. (Make it Passive)
 - Take regular exercise. (Make it Assertive)
 - Reading more, you can learn more. (Make it Compound)
6. Fill in the gaps with the correct form of verbs. 5
- Trees contribute a lot to maintain ecological balance. A month-long tree fair (a)..... (begin) in Kurigram stadium recently. It (b).....(organize) by the local administration and the department of social Forestation. The fair (c).....(inaugurate) yesterday by the DC of kurigram. It was also addressed by other speakers who (d).....(emphasize)the need for planting more trees. The chief guest in his speech (e).....(attach) importance to planting trees so that we can save our environment from other effects of climate change and natural disasters.
7. Fill in the blanks with suitable article. Use cross (X) mark where an article is not necessary. 5
- (a).....idle man and (b).....active man cannot be equal. We know (c)..... story of an ant and (d).....grasshopper. An ant was industrious .On the other hand, the grasshopper was really (e).....lazy.
8. Change the narrative style by using in direct speech. 5
- He said, "I can chop some wood today". I said, "But I have a boy coming from the orphanage". "I am the boy", "You? But you're small". "Size doesn't matter chopping wood", he said.

Part C: Writing (30 marks)

- Write a paragraph on 'Food Adulteration' or 'Dengue Fever'. 8
- Write an application to the Principal/Head of your college/institute for the arrangement of study tour. 8
- Write a report on the unchanged price hike of the essential commodities. 7
- Make a dialogue between two friends on the '2022 FIFA World Cup'. 7